

# Matter-Antimatter Asymmetry at LHCb

## Third Year Lab Report

Damian French

Department of Physics and Astronomy, University of Manchester

(Experiment performed in collaboration with George Jones)

(Dated: May 1, 2026)

The goal of this experiment was to calculate the global matter-antimatter asymmetry in B meson decays at LHCb due to CP violation, and investigate the local asymmetry using Dalitz plot analysis. Utilising data from the detector, we reconstructed the mass distribution of the B mesons, and by fitting a model determined a global asymmetry of  $-0.044 \pm 0.01(\text{syst}) \pm 0.02(\text{stat})$ . Through removing the background using SPlot and creating significance Dalitz plots, we were able to recognise regions of high asymmetry and supply strong evidence for local asymmetry through the Miranda process.

## 1. INTRODUCTION

The universe is primarily made of matter, but why is there so much more of it than antimatter? This is one of the most fundamental questions of particle physics. For the universe to have such an asymmetry, there are three conditions, called the Sakharov conditions, that need to be fulfilled: Baryon number violation, C and CP violation and interactions out of thermal equilibrium [1].

The first case of CP violation was discovered in 1964 [2], and LHCb began operations in 2008 [3] with the purpose of more precisely measuring the CP violation in B meson decays.

CP violation causes a difference in probability of specific weak decay channels between matter and antimatter, and LHCb was designed to produce B mesons at very high rates to measure it. Through a proton-proton collision, many B mesons are produced, which subsequently decay into many more particles which are logged by the detector.

In this experiment an analysis was performed on these detected decay products from the B mesons, in order to determine the number of  $B^+$  and  $B^-$  mesons produced in the collision. Specifically, the  $B^\pm \rightarrow k^\pm k^\pm k^\mp$  decay branch was analysed. This is then used to determine if there is an asymmetry in their decay rates.

## 2. THEORY

CP-symmetry is made from a combination of both parity and charge conjugation symmetries. For an interaction to conserve charge conjugation, it should occur at the same rate under the charge conjugation operator, which converts particles into their antiparticles. Parity conservation works the same way with the parity operator reflecting the spatial coordinates of a particle.

CP symmetry occurs when there is no change to an interaction under application of both operators. It was hoped that all interactions would be CP-symmetric, meaning matter and antimatter would work the same under reflected spatial coordinates, but this was found to not be the case.

B mesons consist of either an up and anti-bottom quark,

$$B^+ \equiv (u, \bar{b}), \quad (1)$$

or an anti-up and a bottom quark,

$$B^- \equiv (\bar{u}, b). \quad (2)$$

These decay through multiple branches, and each one has multiple paths, but the two important for this experiment are shown in Figure 1 for the  $B^+$  decay, with the  $B^-$  decay being the same but with the charge conjugation operator applied.

The reason for these specific diagrams is that they involve  $b \rightarrow u$  weak decays. The amplitude of this decay contains a complex phase in the CKM matrix, which governs the probability amplitude of weak decays [4]. This complex phase

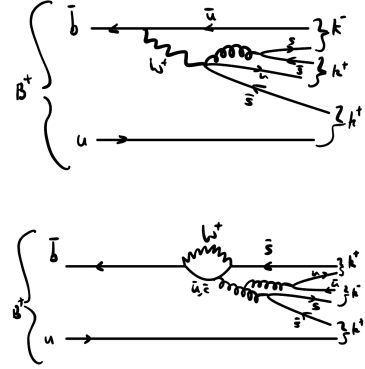


FIG. 1. *top*: tree level diagram for the  $B^+ \rightarrow k^+ k^+ k^-$  branch. *bottom*: Penguin diagram for the  $B^+ \rightarrow k^+ k^+ k^-$  decay, which is suppressed compared to the tree level as it is a higher-order process.

means that under charge conjugation operations the probability changes due to a flipping of the weak phase, producing interference between the two diagrams which causes an asymmetry in decay rates through this channel.

Globally, the asymmetry is caused by both weak and strong phase difference, whilst locally it is the strong phase that changes across small regions causing non-uniform asymmetry across a Dalitz plot.

## 3. METHODOLOGY AND ANALYSIS

In order to determine the asymmetry in decay rates, the decayed B mesons need to be reconstructed from the detected kaons. We can then fit a function to the mass distribution of the B meson and determine how many were present. By doing this for both  $B^+$  and  $B^-$  mesons, we can determine an asymmetry in their populations, and from this determine the asymmetry in their branching fractions.

### 3.1. Reconstruction Mass Distributions

For each B meson decay, there are 3 subsequent kaon tracks detected. Using

$$m^2 = \left( \sum_i E_i \right)^2 - \left( \sum_i p_i \right)^2, \quad (3)$$

we can determine the mass of each B meson, and produce a mass distribution for all the decays detected.

B mesons can also decay into intermediary neutral particles before subsequently decaying into kaons. For any 3 kaons, there are 2 possible ways to make the neutral particle, and one of these will be more massive than the other. We can therefore

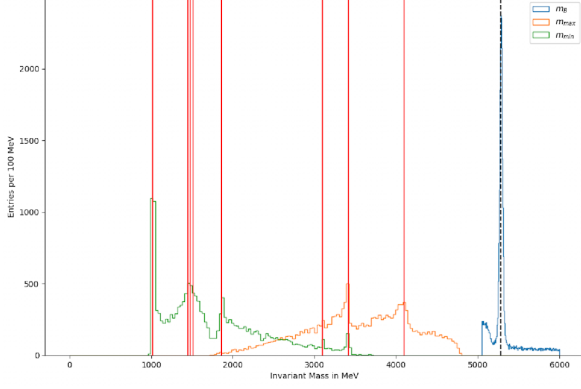


FIG. 2. Figure displaying the mass distributions for the reconstructed B mesons, as well as the possible intermediary particles [5]. The red lines represent peaks where real particles exist. The first is a  $\phi$  meson, the group of 3 is as collection of light mesons, the 5th line is a  $D^0$  meson, the next a  $J/\psi$  meson, followed by a  $\chi_{c0}$  meson and then other charmonious mesons.

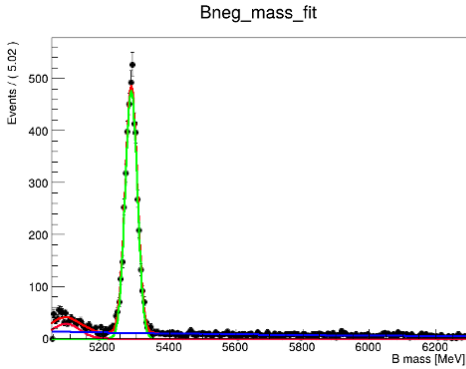


FIG. 3. Figure showing 2 Gaussians and an exponential used to fit the B meson mass distribution (darker red). The exponential function (blue) models the combinatorial background, which is due to other interactions producing particles other than the kaons. One Gaussian models the B meson mass peak itself (green), whilst the other accounts for the misidentification bump on the LHS (red).

use Equation 3 to generate mass distributions for the two possible intermediary particles, as shown in Figure 2. The intermediary particles were filtered so that their 3-body invariant mass given by Equation 3 was within the known width of the B meson mass peak. As explained earlier, the  $b \rightarrow u$  decays contribute to the asymmetry as they violate CP symmetry.  $D^0$  and  $J/\psi$  mesons are produced through the  $b \rightarrow c$  weak decays and so don't contribute. We therefore remove these peaks from the mass distributions in subsequent analysis.

### 3.2. Fitting B Meson Mass Distributions

Once the B meson mass distribution was determined, it was fit using 2 Gaussians and an exponential as shown in Figure 3. This was done using the RooFit library [6], which performs a maximum likelihood fit. This is preferable to a  $\chi^2$  fit in the low-statistics regime, since  $\chi^2$  minimisation assumes Gaussian errors on bin counts, which only holds when  $N \gg 1$ . As can be seen in Figure 3, many bins contain fewer than 10 events, making the Poisson nature of the counting statistics non-negligible. As can be seen in Figure 2, there is a small bump on the left-hand side of the B meson mass distribution. This is caused by a misidentification by the detector of a 4-body decay of  $kkk\pi$  as a 3-body decay, and is accounted for by one of the Gaussians. The other Gaussian models the B

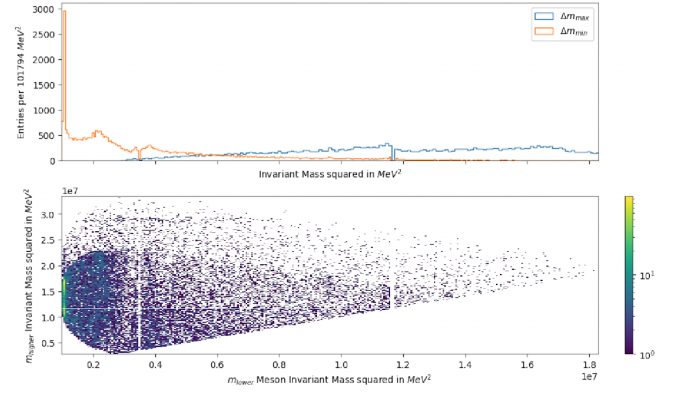


FIG. 4. Dalitz plot of intermediary particles with background included. (top): 1D mass distributions plotted on the same axis. (bottom): 2D mass<sup>2</sup> histogram (Dalitz plot) of the intermediate particles. The colour represents the number of datapoints in each bin. It can be seen that there are gaps where the  $D^0$  and  $J/\psi$  mesons have been removed.

mass peak itself and the exponential models background interactions.

From the fit to the B meson peak, which is called the signal, we could then calculate the global asymmetry using

$$A = \frac{N^- - N^+}{N^- + N^+}. \quad (4)$$

### 3.3. Removing Background

The intermediate mass distributions can be plotted together on a 2D histogram to produce a Dalitz plot. This is a plot showing the mass squared combinations of pairs from a three-body decay. The distribution of points shows resonances contributing to the decay, and the Dalitz plot of the intermediary particles can be seen in Figure 4. Figure 4 shows that there is a lot of background data in our Dalitz plot, since there is not a clean line describing the boundary of physically allowed decays as we would expect. To remove this background we employed a statistical subtraction method called SPlot [7]. Using our fit for the B meson mass each event is assigned a statistical weight based on the how likely each event is to belong to the signal PDF, described as

$${}_s\mathcal{P}_n(y_e) = \frac{\sum_{j=1}^{N_s} \mathbf{V}_{nj} f_j(y_e)}{\sum_{k=1}^{N_s} N_k f_k(y_e)}, \quad (5)$$

where  $N_s$  is the number of PDFs we have in our model,  $\mathbf{V}_{nj}$  is the covariance matrix of each distribution and the signal PDF and  $f_i$  is the current PDF.

The Dalitz plot after SPlot has been applied can be seen in Figure 5, and shows that almost all the background has been removed whilst keeping the desired three-body decays intact.

### 3.4. Local Asymmetry Analysis

With the background removed, We could begin to analyse the local asymmetry in B meson decays. By applying Equation 4 to each bin, we calculated the local asymmetry in each bin. An extra  $\alpha$  coefficient was also used for the  $N^+$  terms, which is defined as

$$\alpha = \frac{N^-}{N^+}, \quad (6)$$

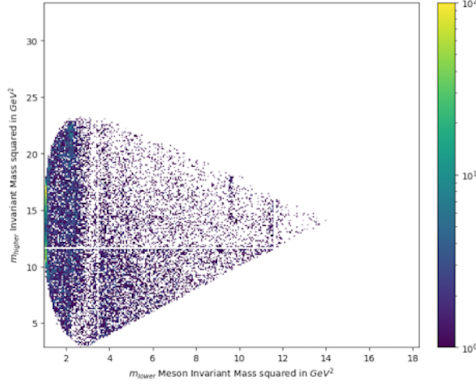


FIG. 5. 2D mass<sup>2</sup> histogram (Dalitz plot) of the intermediate particles in B meson decay with background removed using SPlot. The colour represents the number of datapoints in each bin.

and normalises for the global asymmetry, leaving only the local asymmetry present.

The significance of the asymmetry, defined as

$$\Sigma = \frac{N^- - \alpha N^+}{\sqrt{N^- + \alpha^2 N^+}}, \quad (7)$$

was plotted in Figure 6, and describes the statistical significance between  $\alpha N^+$  and  $N^-$  in each bin. This is useful for testing against a null hypothesis of no asymmetry. The bin

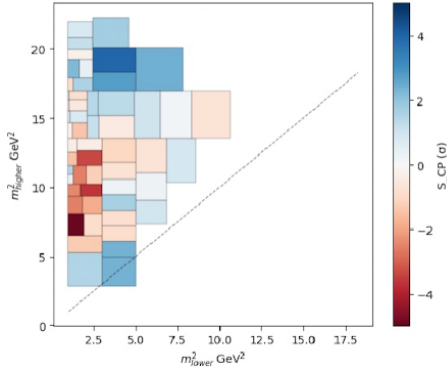


FIG. 6. Dalitz plot with Significance shown as the colour of each bin. The bin widths are variable such that each bin contains the same number of data points, in this case 500. The grey dotted line is the lower boundary of physical 3-body decays.

widths were allowed to vary in this Dalitz plot so that the same number of data points was in each bin. This not only allows for a better visualisation of where the asymmetry is located on the plot, but also allows for a more rigorous analysis. This is because each bin is now in the Gaussian limit, which means they can be tested against normality.

A method called the Miranda process [8] was used to analyse to what extent real local asymmetry caused by CP violation was present. By assuming a null hypothesis of a normal

distribution  $N(0,1)$ , we tested at a 0.05 level of significance whether we had evidence to reject that model for our significance distribution. We found a p-value of 0.09, meaning we did not have sufficient evidence to reject the null hypothesis. The mean was found to be 0.053, and the standard deviation 1.131.

#### 4. DISCUSSION

From our model used to fit to the B meson mass distribution, we found the amplitudes of the signal PDFs for both the  $B^+$  and  $B^-$  mesons. Using Equation 4, we calculated a global asymmetry of -0.044, which is within  $1\sigma$  of the agreed value [9]. The systematic uncertainty in the detector was accounted for by using measurements taken with both spin up and spin down magnets and averaging the two. The systematic uncertainty in the ROOFIT method was calculated by trying multiple models and fits and using a weighted average of their uncertainties weighted with how far their  $\chi_R^2$  was from 1. This resulted in a total systematic uncertainty of  $\pm 0.01$ . By propagating errors on Equation 4, we calculated a statistical uncertainty of  $\pm 0.02$ .

SPlot requires that the control and discriminatory variables are uncorrelated, but upon testing we found a correlation coefficient of 0.2. This meant that there is some bias to the results of our local asymmetry, meaning that any calculated asymmetry values would have underestimated uncertainties.

Although the p-value for the Miranda process was not enough to reject  $H_0$ , the trade off in having enough data points in each bin is that there are fewer bins, and this increases the variance in our p-value, making it a less important statistic. The fact that the standard deviation is greater than 1 suggests our distribution is broader than expected which implies genuine local asymmetry. This combined with the fact that the mean is close to 0 implies that although there is local asymmetry, it is changing sign in different regions. This is expected as the strong phase difference between the two diagrams in Figure 1 should be constructively and destructively interfering in different regions. This is reinforced by Figure 6 which clearly shows regions of both positive and negative significance.

A remaining source of uncertainty is detector-related asymmetry, since differences in detection efficiency between  $K^+$  and  $K^-$  tracks could mimic a genuine CP asymmetry. This can be accounted for by analysing  $b \rightarrow c$  decay channels, which to a good approximation are CP conserving, meaning any asymmetry observed there would be attributable to detector effects and could be subtracted from our result. This correction was not performed here but could be explored in future analysis.

#### 5. CONCLUSION

This experiment attempted to quantify the global asymmetry in  $B^+$  and  $B^-$  decays due to CP violation. We found evidence of global asymmetry with a value of  $A = 0.044 \pm 0.01(\text{syst}) \pm 0.02(\text{stat})$ , which is within 1 standard deviation of the value provided by LHCb [9]. We also found evidence consistent with local asymmetry, which could be seen qualitatively in the significance Dalitz plot and also tested for using the Miranda process.

[1] A. D. Sakharov, Pisma Zh. Eksp. Teor. Fiz. **5**, 32 (1967).  
 [2] J. H. Christenson *et al.*, Phys. Rev. Lett. **13**, 138 (1964).  
 [3] LHCb Collaboration, Journal of Instrumentation **3** (08), S08005.  
 [4] M. Kobayashi and T. Maskawa, Prog. Theor. Phys. **49**, 652 (1973).  
 [5] Particle Data Group, Phys. Rev. D **110**, 030001 (2024).  
 [6] W. Verkerke and D. Kirkby, arXiv (2003), physics/0306116.

[7] M. Pivk and F. Le Diberder, Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment **555**, 356 (2005).  
 [8] I. Bediaga, I. I. Bigi, A. Gomes, G. Guerrer, J. Miranda, and A. C. dos Reis, Physical Review D **80**, 10.1103/physrevd.80.096006 (2009).  
 [9] LHCb Collaboration, Phys. Rev. Lett. **111**, 101801 (2013).